

Mount a Motor and Coupling SOP (1x Episode: 1 Motor and Coupling)

One episode mounts one motor and joins it to one driven shaft with one coupling. The table begins with the **base plate** on the assembly spot, just right of the center of the table. The plate carries the **motor bracket** at its right end and the **bearing block** at its left end. The **driven shaft** sticks out of the bearing block and points right, toward the bracket. A **parts tray** holds the motor, its two wing bolts, and the coupling. The **gap gauge** stands in its **holder** at the front-left.

The order never changes: set the motor on the bracket, bolt it down, slide the coupling onto both shafts, set the gap with the gauge, tighten the two set screws, then hand-spin the drive. No bolt is tightened until both bolts are started by hand. The coupling is never touched by a set screw until the gauge has set the gap. The episode does not end until the drive has turned two full turns with the driven shaft turning with it.

The right gripper carries the motor and the coupling, seats the motor, turns both bolts, slides the coupling, turns both set screws, and drives the hand-spin. The left gripper presses the base plate flat against the table, and holds the gap gauge against the motor face while the coupling is slid. The left gripper works from the left of the plate and never reaches across it. The right gripper works the parts tray, the motor, the bolts, the coupling, and the set screws.

The table is set up in one of three ways. Only the parts tray moves; the base plate on its assembly spot and the gauge holder are in the same place in all three.

- **Config M:** the parts tray is at the front-center, in front of the base plate.
- **Config R1:** the parts tray is at the front-right.
- **Config R2:** the parts tray is at the back-right, behind the motor bracket.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the parts tray starts at the front-center (**Config M**), the front-right (**Config R1**) or the back-right (**Config R2**). One config per episode, chosen before recording and never changed mid-episode.
- **No left zone:** the left gripper is on the plate edge or the gauge for the whole episode and the left side of the table is kept clear for it, so the parts tray never starts on the left and there is no Config L.
- **Same-side rule:** the **right gripper** takes every part out of the parts tray in all three configs. The left gripper never reaches for a part, and no arm reaches across the table.

- **Fixed roles:** the left gripper holds the plate and then the gauge, the right gripper seats, bolts, couples, gaps, tightens, and spins, and the order of Steps 1–8 is the same in all three configs.

Setup

Complete both checklists before recording starts.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table: the base plate with the motor bracket and the bearing block, the parts tray in the start zone for this episode's config, and the gauge holder at the front-left.
3. The base plate is visible from above and from the front, so both bolt heads, the gap between the motor face and the coupling, both set screws, and both witness lines can be seen.
4. Both arms are at home with their grippers open.
5. The table is clear of everything except the base plate, the parts tray, and the gauge holder.
6. The right arm reaches the parts tray (front-center in Config M, front-right in Config R1, back-right in Config R2), the bracket, both bolts, the coupling, and both set screws without stretching. The left arm reaches the left edge of the base plate and the gauge holder without stretching.

Materials checklist

1. One **base plate** lies on the **assembly spot**, just right of the center of the table, square to the front edge and flat on the table.
2. The **motor bracket** is fixed at the right end of the plate. It has a flat **bracket pad** for the motor foot and two open **bolt holes**, one at the front and one at the back.
3. The **bearing block** is fixed at the left end of the plate. The **driven shaft** sticks out of it and points right, toward the bracket. The shaft carries a painted **hub mark** on its collar, next to the bearing block.
4. One **motor** lies on its side in the **parts tray**, with its two **wing bolts** loose in the tray beside it. The motor shaft points out of the motor face; the **spin knob** with its own painted mark sits on the back of the motor. The tray stands in the start zone for this episode's config:
 - **Config M:** front-center, in front of the base plate, clear of the gauge holder
 - **Config R1:** front-right

- **Config R2:** back-right, behind the motor bracket, clear of the plate
5. One **coupling** lies in the same parts tray. It is a plain sleeve with one **set screw** near each end, both backed off far enough that the coupling slides freely along a shaft.
 6. Each shaft carries a painted **witness line** around it, part way along. The line marks how far the shaft must go into the coupling.
 7. One **gap gauge** stands in the **gauge holder** at the front-left.
 8. Both shafts are clean and dry, the coupling bore is clean, and neither shaft is bent, burred, or scored.
 9. Keep the left side of the table clear. The left gripper works from there onto the plate and the gauge holder.
 10. Before collection, confirm by hand that the motor foot sits flat on the bracket pad with both bolt holes lined up, that both wing bolts turn all the way down by hand, that the coupling slides onto both shafts without force, that the gauge fits between the motor face and the coupling with the coupling still covering both witness lines, and that turning the spin knob turns the driven shaft with it.

Workspace layout

- **Assembly spot:** just right of the center of the table — the base plate stays here all episode
- **Right end of the plate:** the motor bracket, its pad, and its two bolt holes
- **Left end of the plate:** the bearing block and the driven shaft with its hub mark
- **Middle of the plate:** the space the coupling ends up in, between the two shafts
- **Start zone:** the parts tray with the motor, the two wing bolts, and the coupling — front-center (Config M), front-right (Config R1), or back-right (Config R2)
- **Front-left:** the gauge holder with the gap gauge
- **Left side:** kept clear, so the left gripper can come in onto the plate's left edge and the gauge holder

Arm assignments

- **Left gripper:** presses the base plate flat against the table by its left edge while the motor is seated, bolted, and coupled, and holds the gap gauge flat against the motor face while the coupling is slid to the gauge. The same in all three configs.
- **Right gripper:** carries the motor, the bolts, and the coupling from the parts tray — front-center in Config M, front-right in Config R1, back-right in Config R2 — seats the motor on the bracket pad, turns both wing bolts, slides the coupling along the shafts, turns both set screws, and drives the hand-spin on the spin knob.

Vocabulary

- **Base plate:** the flat plate the drive is built on. It stays on the assembly spot for the whole episode and is never lifted.
- **Motor bracket:** the fixed stand at the right end of the plate that the motor bolts onto.
- **Bracket pad:** the flat top face of the bracket that the motor foot sits on.
- **Bolt hole:** either of the two open holes through the bracket pad, one at the front and one at the back.
- **Wing bolt:** a bolt with a flat head the right gripper can pinch and turn. There are two.
- **Wing:** the flat head of a wing bolt or a set screw, which the right gripper pinches to turn it.
- **Motor:** the part that is bolted to the bracket. Its **motor foot** is the flat base that sits on the pad.
- **Motor face:** the end face of the motor that the motor shaft comes out of. It faces left, toward the bearing block.
- **Motor shaft:** the shaft sticking out of the motor face, pointing left.
- **Spin knob:** the knob on the back of the motor, with a painted mark on it. Pushing the knob sideways turns the motor shaft.
- **Bearing block:** the fixed block at the left end of the plate that carries the driven shaft.
- **Driven shaft:** the shaft sticking out of the bearing block, pointing right.
- **Hub mark:** the painted mark on the driven shaft's collar, beside the bearing block. It shows whether the driven shaft is turning.
- **Coupling:** the plain sleeve that slides over both shafts and joins them.
- **Bore:** the hole through the middle of the coupling that the shafts go into.
- **Set screw:** either of the two screws in the coupling, one near each end. The one over the motor shaft is the **motor-side set screw**; the one over the driven shaft is the **driven-side set screw**.
- **Witness line:** the painted line around each shaft. If the line is hidden by the coupling, that shaft is far enough into the bore. If the line shows, it is not.
- **Gap gauge:** the flat tab kept in the gauge holder. Standing it against the motor face sets how far the coupling sits from the motor.
- **Gauge holder:** the stand at the front-left that the gap gauge lives in.
- **Start zone:** where the parts tray stands at the start of the episode — front-center (**Config M**), front-right (**Config R1**), or back-right (**Config R2**). One per episode, chosen before recording and never changed mid-episode.
- **Flat against:** the two faces touch all over, with no daylight showing between them at any edge.
- **Started by hand:** the bolt has been turned in far enough to stand in its hole on its own, but is not tight yet.

- **Just fits:** the gauge is held between the motor face and the coupling end with both faces touching it, and the gauge is not bent, squeezed, or pushed out of line.
- **Stops turning:** the wing will not turn any further with a normal pinch and pull.
- **Full turn:** the spin knob has gone all the way around, so its mark comes back to where it started.
- **Turns with it:** the hub mark on the driven shaft moves whenever the spin knob moves.
- **Slip:** the spin knob turns but the hub mark stops or falls behind.
- **Bind:** the spin knob will not turn any further when it is pushed.
- **Walk:** the coupling slides along the shafts by itself while the drive is turned, so the gap at the motor face opens or closes.

Steps

Run Steps 1–7 in order on the one motor and coupling, then end the episode with Step 8.

Steps 2, 3.1 and 4 depend on where the parts tray is: the **right gripper** takes every part from it in all three configs, so only the reach changes — from the front-center in Config M, from the front-right in Config R1, from the back-right in Config R2. Step 7 has one line for Config R2, where the empty tray lies behind the motor. Every other line is the same in all three configs.

Step 1: Hold the base plate down

Goal: the base plate is pinned flat on the assembly spot and cannot slide while the motor goes on.

- With the **left gripper**, press down on the **left edge** of the base plate and hold it against the table.
- Keep the **left gripper** there through Steps 2, 3 and 4.

Check: the plate is flat on the table, square to the front edge, and does not slide when the left gripper presses. If the plate is crooked, straighten it with the left gripper first, then press it down.

Expected state: the plate is held, the bracket pad is empty, and both shafts are bare.

Step 2: Set the motor on the bracket

Goal: the motor sits flat on the bracket pad with its shaft pointing at the driven shaft and both bolt holes lined up.

- With the **left gripper**, keep pressing the base plate down.

Look where the parts tray is before reaching for the motor.

- **IF the parts tray is at the front-center (Config M):** with the **right gripper**, pinch the motor by its **body**, lift it out of the parts tray, and carry it back and to the right, low over the front of the plate, to the bracket.

- **IF the parts tray is at the front-right (Config R1):** with the **right gripper**, pinch the motor by its **body**, lift it out of the parts tray, and carry it straight back to the bracket.
- **IF the parts tray is at the back-right (Config R2):** with the **right gripper**, pinch the motor by its **body**, lift it out of the parts tray, and carry it straight forward to the bracket.

Then, in all three:

- Carry the motor in one go, straight from the tray to the bracket. Do not set it down on the plate or the table on the way.
- Do not pick the motor up by its shaft, and do not let the shaft take the weight.
- With the **right gripper**, lower the motor onto the **bracket pad** so the **motor face** points left, toward the bearing block, and the motor foot sits down on the pad.
- With the **right gripper**, nudge the motor until the two holes in its foot line up over the two **bolt holes** in the pad.
- Let go once the motor sits still on the pad.

Expected state: the motor sits on the bracket pad, its shaft points left at the driven shaft, and both bolt holes show clear through the foot.

Check: the motor foot is **flat against** the pad, the two shafts point straight at each other, and both bolt holes line up. If the foot rocks or a hole is covered, lift the motor with the right gripper, set it down again, and line it up.

Step 3: Bolt the motor to the bracket

Goal: both wing bolts are in and tight, and the motor cannot move on the pad.

- With the **left gripper**, keep pressing the base plate down.
- Start both bolts before tightening either one.

3.1 Start both bolts by hand

- **IF Config M:** reach forward and left to the parts tray for each bolt. **IF Config R1:** reach forward. **IF Config R2:** reach back, over the bracket.
- With the **right gripper**, pinch the **wing** of one bolt, lift it out of the parts tray, and stand it in the **back bolt hole**.
- With the **right gripper**, turn the bolt **clockwise** until it stands in the hole on its own.
- Turn only while the bolt stays straight up. If it leans or will not turn, back it out counter-clockwise with the **right gripper**, stand it straight, and start again.
- With the **right gripper**, do the same with the second bolt in the **front bolt hole**.

3.2 Tighten the bolts

- With the **right gripper**, pinch the wing of the **back bolt** and turn it clockwise a quarter turn.
- Open the **right gripper**, take the wing of the **front bolt**, and turn that one a quarter turn.

- Go back and forth between the two bolts, a quarter turn each time, until each bolt **stops turning**.
- Do not turn more than a quarter turn in one grip, and do not keep forcing a bolt once it stops.

Expected state: both bolts are down on the foot, the motor sits flat on the pad, and the motor does not move when the right gripper lets go.

Check: both wings are down against the motor foot and neither bolt turns any further. If the motor still rocks or slides on the pad, take each wing again with the right gripper and turn a further quarter turn each, back and forth, until both stop.

Step 4: Slide the coupling onto both shafts

Goal: the coupling is on both shafts, pushed home to the bearing block, with the set screws still loose.

- With the **left gripper**, keep pressing the base plate down.
- **IF Config M:** reach forward to the parts tray for the coupling. **IF Config R1:** reach forward and right. **IF Config R2:** reach back and right, over the bracket.
- With the **right gripper**, pinch the coupling by its outside and lift it out of the parts tray.
- Carry the coupling in one go, straight from the tray to the shafts. Do not set it down on the way.
- With the **right gripper**, hold the coupling in line with the two shafts and slide it **left** onto the **motor shaft** first.
- Keep sliding it left until the far end of the bore takes the **driven shaft** as well.
- Push the coupling left along both shafts until it comes up against the **bearing block**.
- Slide the coupling straight along the shafts only. Do not rock it, twist it onto a shaft, or pry it with the gripper tip.
- Do not touch either set screw yet.

Expected state: the coupling covers the end of both shafts, sits against the bearing block, and still slides when pushed.

Check: the coupling is on both shafts and both set screws are still loose. If the coupling will not go on straight, pull it back off with the right gripper, line it up with the shafts, and slide it on again. If it still will not go on, end the episode at Step 8 with the coupling in the tray and log the shafts for a station check.

Step 5: Set the gap with the gauge

Goal: the coupling sits one gauge away from the motor face, with both shafts still far enough into the bore.

5.1 Hold the gauge on the motor face

- With the **left gripper**, let go of the plate edge and pinch the **gap gauge** in the gauge holder.
- With the **left gripper**, lift the gauge out and bring it in over the plate.
- With the **left gripper**, stand the gauge **flat against** the **motor face**, beside the motor shaft, and hold it there.

5.2 Slide the coupling to the gauge

- With the **right gripper**, pinch the coupling by its outside and slide it **right**, back toward the motor, until its right end touches the gauge.
- Stop as soon as the coupling touches the gauge. Do not push the coupling into the gauge.
- Let go with the **right gripper** and leave the coupling where it is.

5.3 Take the gauge out

- With the **left gripper**, draw the gauge straight **down and out** of the gap, without dragging it against the coupling.
- With the **left gripper**, put the gauge back in the gauge holder.
- With the **left gripper**, come back to the **left edge** of the base plate and press it down again. Keep it there through Steps 6 and 7.

Expected state: the coupling stands clear of the motor face, the gauge is back in its holder, and neither witness line shows.

Check: the coupling did not move when the gauge came out, and **no witness line** shows on either shaft. If the coupling moved, hold the gauge on the motor face again with the left gripper and slide the coupling back to it with the right gripper. If a witness line shows on the driven shaft, the coupling has come too far right: end the episode at Step 8 and log the shaft spacing for a station check.

Step 6: Tighten the two set screws

Goal: both set screws are tight, so the coupling holds on both shafts.

- With the **left gripper**, keep pressing the base plate down.
- Tighten the **motor-side set screw** first, then the **driven-side set screw**.
- With the **right gripper**, pinch the wing of the **motor-side set screw** and turn it **clockwise** a quarter turn.
- Open the **right gripper**, take the wing again at the start of a fresh quarter turn, and turn again.
- Repeat these quarter turns until the screw **stops turning**.
- With the **right gripper**, do the same on the **driven-side set screw**, until it stops turning.
- Do not turn more than a quarter turn in one grip, and do not keep forcing a screw once it stops.

- Do not push or pull the coupling along the shafts while turning a screw.

Expected state: both set screws are down, the coupling does not slide when the right gripper nudges it, and the gap at the motor face is still open.

Check: both wings stop turning, the coupling still stands clear of the motor face, and no witness line shows. If the coupling moved while the screws were turned, back both screws off counter-clockwise with the right gripper until the coupling slides again, then go back to Step 5.

Step 7: Hand-spin the drive

Goal: the spin knob makes two full turns, the driven shaft turns with it, and the coupling stays put.

- With the **left gripper**, keep pressing the base plate down.
- With the **right gripper**, close the gripper and set its tip against the **spin knob**.
- **IF Config R2:** the empty parts tray lies behind the motor; bring the tip in to the knob from above, without touching the tray.
- Push the knob sideways to turn it. Lift the tip clear, set it back on the knob, and push again.
- Keep pushing until the spin knob has made **two full turns**, so its mark comes back to where it started twice.
- While the knob turns, watch the **hub mark** on the driven shaft. It must move every time the knob moves.
- While the knob turns, watch the coupling. It must stay where it is and not **walk** along the shafts.
- Drive from the spin knob only. Do not push on the coupling, do not turn the driven shaft, and do not grab a shaft and twist it.
- If the knob makes both turns, the hub mark turns with it, and the coupling stays put, go to Step 8.
- If the hub mark stops or falls behind, there is **slip**: go back to Step 6, turn both set screws further in quarter turns, then run Step 7 again from the start.
- If the knob will not turn any further, there is **bind**, and if the coupling walks, the gap is wrong: back both set screws off with the **right gripper** until the coupling slides, go back to Step 5, then run Steps 6 and 7 again from the start.
- If the same fault comes back after two corrections, stop, end the episode at Step 8 as it stands, and log the drive for a station check.

Check: the spin knob made two full turns, the hub mark moved with it every time, the coupling is still clear of the motor face, and the motor is still tight on the bracket.

Step 8: End the episode

Goal: recording ends with the motor bolted down, the coupling gapped, both set screws tight, and the drive spun.

1. Confirm the motor sits flat on the bracket pad with both bolts tight, the coupling stands clear of the motor face with no witness line showing, both set screws are down, the gauge is back in its holder, and the base plate is still square on the assembly spot.
2. Return both arms home with grippers open. Homing is the last thing the arms do.
3. Stop recording.

After the episode: reset the workspace

All reset work happens with recording off.

1. With recording off, back both set screws off by hand until the coupling slides freely on both shafts.
2. Pull the coupling off the driven shaft, then off the motor shaft, and lay it in the parts tray.
3. Back both wing bolts out of the bracket by hand and lay them in the parts tray.
4. Lift the motor off the bracket pad and lay it on its side in the parts tray.
5. Set the base plate square on the assembly spot, flat against the table, and put the gap gauge back in its holder. Set the parts tray in the start zone for the next episode's config — front-center (Config M), front-right (Config R1), or back-right (Config R2).
6. Wipe grit, dust, and metal shavings off both shafts, out of the coupling bore, and off the bracket pad.
7. Inspect the parts. Replace the coupling if the bore is scored or a set screw no longer bites. Replace a wing bolt if its thread no longer pulls the motor down. Replace the gap gauge if it is bent. Tighten the bracket or the bearing block if either is loose on the plate, and replace a shaft that is bent or burred.
8. Run both Setup checklists before the next episode.

SOP violations

Things that break this SOP and that reviewers look for in the side-by-side review tool.

How to record a violation in review

For every violation seen in a recorded episode, record:

- the **start timestamp** in the video;
- the **violation name** from the list below; and
- the **SOP rule broken**, including the step number.

The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. An episode may contain multiple violations; tag each separately. Retain the episode in training data with its violation tags. Do not delete a recorded episode solely because it contains a violation.

Violations

Note on the start position: the violations below were written for Config R1 (parts tray at the front-right). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table — the parts tray is not in the start zone for the config; the left gripper reaches for a part, or a gripper reaches across the table for the parts tray; or the wrong IF line is followed.
- **SOP rule broken:** the start position and the same-side rule (the parts tray starts at the front-center, the front-right, or the back-right, and the right gripper takes every part from it; no arm reaches across the table; the IF line followed is the one for the config on the table).
- **Coaching note:** look where the parts tray is before the first reach, then follow that config's IF lines through Steps 2, 3, 4 and 7.

Violation: Base plate not held

- **Visible cue:** the left gripper is off the left edge of the base plate while the right gripper seats the motor, turns a bolt, slides the coupling, or turns a set screw, and the plate slides, lifts, or is chased across the spot.
- **SOP rule broken:** Steps 1, 3, 4, 6 and 7 (the left gripper presses the base plate down by its left edge and holds it there whenever it is not holding the gauge).
- **Coaching note:** left gripper on the plate first, and back on the plate as soon as the gauge is stowed.

Violation: Base plate moved off the assembly spot

- **Visible cue:** the base plate is lifted, turned, dragged, or ends the episode out of square with the front edge.
- **SOP rule broken:** Steps 1–7 (the base plate stays flat and square on the assembly spot for the whole episode and is never lifted).
- **Coaching note:** bring the parts to the plate; never move the plate to the parts.

Violation: Motor mishandled on the way in

- **Visible cue:** the right gripper picks the motor up by its shaft, lets the shaft take the weight, or sets the motor down on the plate or the table on the way from the parts tray.
- **SOP rule broken:** Step 2 (pinch the motor by its body and carry it in one go, straight from the tray to the bracket).
- **Coaching note:** one lift, by the body, straight to the pad. The shaft is not a handle.

Violation: Motor seated wrong on the bracket

- **Visible cue:** the motor foot rocks or sits part way off the bracket pad, the motor face points somewhere other than at the driven shaft, or a bolt hole is still covered when Step 3 starts.
- **SOP rule broken:** Step 2 (the motor foot is flat against the pad, the shaft points at the driven shaft, and both bolt holes line up before any bolt goes in).
- **Coaching note:** flat, pointing left, holes showing. Lift and reset rather than nudging a rocking motor.

Violation: Bolts not both started before tightening

- **Visible cue:** the right gripper tightens the first bolt down before the second is standing in its hole, or a bolt is turned while leaning in the hole instead of being backed out and stood straight.
- **SOP rule broken:** Step 3 (start both bolts by hand in 3.1, standing straight, before tightening either in 3.2).
- **Coaching note:** both bolts in and standing first. A leaning bolt comes back out, it does not get forced.

Violation: Bolts tightened out of order

- **Visible cue:** one bolt is run all the way down before the other is touched, instead of the right gripper going back and forth a quarter turn at a time.
- **SOP rule broken:** Step 3 (back bolt, front bolt, a quarter turn each, back and forth until both stop).
- **Coaching note:** alternate. One bolt run down first pulls the motor crooked on the pad.

Violation: Bolt left loose

- **Visible cue:** the episode goes on with a wing standing off the motor foot, or the motor still rocks or slides on the pad when the right gripper lets go.
- **SOP rule broken:** Step 3 (turn each bolt in quarter turns until it stops turning, and the motor does not move when the right gripper lets go).
- **Coaching note:** quarter turn, swap bolts, again, until both stop. Then push the motor to prove it.

Violation: Coupling forced or put on out of order

- **Visible cue:** the coupling goes onto the driven shaft before the motor shaft, or the right gripper rocks it, twists it onto a shaft, pries it with the gripper tip, or pushes it on hard instead of sliding it straight.
- **SOP rule broken:** Step 4 (slide the coupling left onto the motor shaft first, then onto the driven shaft, straight along the shafts, without rocking, twisting, or prying).
- **Coaching note:** motor shaft first, straight slide. If it needs force, pull it off and line it up again.

Violation: Shaft not far enough into the coupling

- **Visible cue:** a witness line still shows on the motor shaft or the driven shaft, and the episode goes on to the set screws or the ending.
- **SOP rule broken:** Steps 4 and 5 (push the coupling home to the bearing block, and end Step 5 with no witness line showing on either shaft).
- **Coaching note:** look along both shafts for paint. Any line you can see means the shaft is too shallow.

Violation: Gap gauge skipped

- **Visible cue:** the coupling is set by eye and the set screws are turned with the gap gauge never leaving its holder.
- **SOP rule broken:** Step 5 (the left gripper holds the gauge on the motor face and the coupling is slid to it before any set screw is touched).
- **Coaching note:** the gauge sets the gap, every episode. Eyeballing it is not the same job.

Violation: Gauge held wrong

- **Visible cue:** the left gripper holds the gauge tilted, off the motor face, against the coupling instead of the motor, or lets it hang in the gap without touching either face.
- **SOP rule broken:** Step 5 (stand the gauge flat against the motor face, beside the motor shaft, and hold it there).
- **Coaching note:** flat on the motor face first, then bring the coupling to it.

Violation: Coupling pushed into the gauge

- **Visible cue:** the right gripper keeps sliding the coupling after it touches the gauge, and the gauge bends, squeezes out of line, or is pushed back against the motor face.
- **SOP rule broken:** Step 5 (stop as soon as the coupling touches the gauge; do not push the coupling into it).
- **Coaching note:** touch and stop. A squeezed gauge sets a gap that is too small.

Violation: Gauge removed or stowed wrong

- **Visible cue:** the coupling moves as the gauge comes out, the left gripper drags the gauge sideways against the coupling, or the gauge is left on the plate, in the tray, or on the table instead of back in its holder.
- **SOP rule broken:** Step 5 (draw the gauge straight down and out without dragging it, leave the coupling where it is, and put the gauge back in the gauge holder).
- **Coaching note:** straight down and out, then straight back to the holder.

Violation: Set screws tightened out of order

- **Visible cue:** the right gripper turns the driven-side set screw before the motor-side set screw is turned down.
- **SOP rule broken:** Step 6 (motor-side set screw first, driven-side set screw second).
- **Coaching note:** motor side first. It pins the coupling before the far end takes any load.

Violation: Set screw left loose

- **Visible cue:** the episode ends with a set screw never turned, or turned only part way, or the coupling still slides when the right gripper nudges it.
- **SOP rule broken:** Step 6 (turn each set screw clockwise in quarter turns until it stops turning, and the coupling does not slide).
- **Coaching note:** both screws, quarter turns, until each stops. Then nudge the coupling to prove it.

Violation: Wing turned the wrong way or forced

- **Visible cue:** the right gripper turns the wing of a bolt or a set screw counter-clockwise while tightening, twists more than a quarter turn in one grip, or keeps forcing a wing after it has stopped.
- **SOP rule broken:** Steps 3 and 6 (turn every wing bolt and every set screw clockwise, a quarter turn per grip, and stop once it stops turning).
- **Coaching note:** clockwise, small bites, let go and re-grip. Stop when it stops.

Violation: Coupling moved while the screws were turned

- **Visible cue:** the right gripper pushes or pulls the coupling along the shafts during Step 6, or the gap at the motor face closes up or opens out while a screw is being turned.
- **SOP rule broken:** Step 6 (do not push or pull the coupling along the shafts while turning a screw; if it moved, back both screws off and go back to Step 5).
- **Coaching note:** the gap is already set. Turn the screw only, and check the gap after each one.

Violation: Hand-spin skipped or cut short

- **Visible cue:** the episode ends without the right gripper driving the spin knob, or the knob's mark comes back to its start fewer than two times.
- **SOP rule broken:** Step 7 (turn the spin knob two full turns, so its mark comes back to where it started twice).
- **Coaching note:** two full turns of the knob, every episode, before the ending.

Violation: Spin driven from the wrong place

- **Visible cue:** the right gripper pushes on the coupling, turns the driven shaft, or grabs a shaft and twists it instead of pushing the spin knob sideways.
- **SOP rule broken:** Step 7 (set the closed gripper tip against the spin knob and push it sideways; drive from the spin knob only).
- **Coaching note:** knob only, tip on the knob, push sideways.

Violation: Slip, bind or walk ignored

- **Visible cue:** the hub mark stops or falls behind, the spin knob will not turn further, or the coupling slides along the shafts during the spin, and the episode moves on to the ending.
- **SOP rule broken:** Step 7 (on slip go back to Step 6; on bind or walk back the screws off and go back to Step 5).
- **Coaching note:** watch the hub mark and the gap, not just the knob. If either goes wrong, fix and retest.

Violation: Correction not re-tested

- **Visible cue:** the set screws are turned further, or the gap is reset, after a slip, bind, or walk, and the episode ends with no fresh pair of full turns.
- **SOP rule broken:** Step 7 (run Step 7 again from the start after every correction).
- **Coaching note:** every fix earns two more full turns.

Violation: Work done out of order

- **Visible cue:** the steps run out of sequence, for example the coupling goes on before the motor is bolted, a set screw is turned before the gauge has set the gap, or the spin test runs before both set screws are down.
- **SOP rule broken:** Steps 2–7 (seat, bolt, couple, gap, set screws, spin, in that order).
- **Coaching note:** the order is the task. Say the next step's name before you move.

Violation: Wrong arm used

- **Visible cue:** an action assigned to one gripper is done by the other, including the left gripper seating the motor, turning a bolt or a set screw, sliding the coupling, or driving the spin knob, or the right gripper holding the gap gauge.

- **SOP rule broken:** Steps 1–7 (the right gripper seats the motor, turns the bolts and the set screws, slides the coupling, and drives the spin; the left gripper holds the plate, then the gauge).
- **Coaching note:** right gripper does the parts and the screws, left gripper holds the plate and the gauge.

Violation: Part or supply knocked off its spot

- **Visible cue:** the motor, a bolt, the coupling, or the gauge is dropped on the plate, the table, or the floor, or the parts tray or the gauge holder is tipped, pushed off its spot, or spilled.
- **SOP rule broken:** Steps 2–7 (keep the motor, the bolts, the coupling, the gauge, the parts tray, and the gauge holder on their spots through the mount).
- **Coaching note:** work slower and lower over the table, and keep the carry short.

Violation: Wrong episode ending

- **Visible cue:** recording stops before the end state is confirmed, an arm is not home, a gripper is closed, or an arm does something else after homing.
- **SOP rule broken:** Step 8 (confirm the end state, return both arms home with grippers open as their final action, then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Non-violation failures

Failures not caused by how the task was run are system issues. Log and discard the episode rather than tagging them as SOP violations.

- Recording stops or pauses during the episode.
- A camera drops frames or loses its feed.
- An arm or gripper fails, drifts, or reports a motor error.
- A wing bolt's thread is stripped, so a fully turned bolt does not pull the motor down onto the pad.
- A set screw's thread is stripped or its tip is rounded, so a fully turned screw does not hold the coupling.
- A shaft is bent, burred, or scored, so a correctly lined-up coupling will not slide on.
- The coupling bore is worn or oval, so a correctly tightened coupling still slips on the shaft.
- The bracket or the bearing block is loose on the base plate, so the shafts do not line up however the motor is bolted.

Annotation subtasks (from SOP)

1. Press the base plate flat and hold it with the left gripper
2. Carry the motor from the parts tray and set it on the bracket pad

3. Start both wing bolts by hand in the bracket
4. Tighten both wing bolts a quarter turn at a time, back and forth
5. Carry the coupling from the parts tray and slide it onto both shafts
6. Hold the gap gauge on the motor face with the left gripper
7. Slide the coupling to the gauge and take the gauge back to its holder
8. Tighten the motor-side set screw, then the driven-side set screw
9. Push the spin knob two full turns and watch the hub mark
10. Confirm the end state, return both arms home, and end the episode